

Problem 26.5

An annuity pays \$100 at the end of one month. It pays \$110 at the end of the second month. It pays \$120 at the end of the third month. The payments continue to increase by \$10 each month until the last payment is made at the end of the 36th month. Find the present value of the annuity at 9% compounded monthly.

Solution

The nominal interest rate is 9% compounded monthly, so the monthly effective interest rate is

$$j = \frac{0.09}{12} = 0.0075.$$

The payments are

$$100, 110, 120, \dots, 450.$$

This can be written as a level annuity of 90 plus an increasing annuity with payments

$$10, 20, 30, \dots, 360.$$

Therefore, the present value is

$$PV = 90a_{\overline{36}|j} + 10(Ia)_{\overline{36}|j}.$$

For an increasing annuity-immediate,

$$(Ia)_{\overline{n}|} = \frac{\ddot{a}_{\overline{n}|} - nv^n}{i}.$$

Using the monthly rate j , we have

$$(Ia)_{\overline{36}|j} = \frac{\ddot{a}_{\overline{36}|j} - 36(1+j)^{-36}}{j}.$$

Thus,

$$PV = 90a_{\overline{36}|j} + 10 \left(\frac{\ddot{a}_{\overline{36}|j} - 36(1+j)^{-36}}{j} \right).$$

Since

$$a_{\overline{36}|j} = \frac{1 - (1+j)^{-36}}{j} \quad \text{and} \quad \ddot{a}_{\overline{36}|j} = \frac{1 - (1+j)^{-36}}{d},$$

where $d = \frac{j}{1+j}$, substitute $j = 0.0075$:

$$PV = 90 \left(\frac{1 - (1.0075)^{-36}}{0.0075} \right) + 10 \left(\frac{\frac{1 - (1.0075)^{-36}}{0.0075/1.0075} - 36(1.0075)^{-36}}{0.0075} \right).$$

Therefore,

$$PV = 8394.60401.$$

Thus, the present value is

$$\boxed{8394.60}.$$